

Mahta Ghafarianghadim

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Georgia State University, Atlanta, GA 30303

EDUCATION

Ph.D. in Economics (Expected 2026), Georgia State University, Atlanta, GA

Chair: Professor Garth Heutel | Committee: Professors Tchernis, Marton, Oliva

M.A. in Economics (2025), Georgia State University, Atlanta, GA

M.A. in Energy Economics (2020), Allameh Tabataba'i University, Tehran, Iran

B.A. in Theoretical Economics (2018), Allameh Tabataba'i University, Tehran, Iran

RESEARCH INTERESTS

Environmental Economics, Health Economics, Policy Evaluation, Applied Microeconomics, Causal Inference

WORKING PAPERS

1. "Air Pollution and Cardiovascular Mortality: Evidence from Thermal Inversions in the U.S." Job Market Paper. Work in Progress.

Provides causal estimates of the effect of PM2.5 on cardiovascular mortality at the county-month level using restricted-access NCHS mortality microdata linked to satellite-derived environmental measures. Systematically compares thermal inversion instruments across multiple specifications; model-level instruments at 110 and 320 meters above local ground yield valid estimates while fixed pressure level instruments fail due to terrain coverage problems. Using the preferred specification, a one-unit increase in PM2.5 raises cardiovascular mortality by 3.8 percent at the county-month level. Results are robust to alternative outcome transformations, PM2.5 measures, pollutants, and the pre-pandemic sample.

2. "Mental Health and Functional Consequences of Smoking: A Causal Approach Using Misreporting Corrections." Working Paper.

Estimates the causal impacts of smoking on mental health and functional limitations using BRFSS data and cigarette tax variation, highlighting the importance of accounting for smoking status misreporting in the evaluation of tobacco and public health policies.

3. "Assessing the Causal Impact of Air Pollution on Childcare Outcomes in the U.S." Working Paper. Joint with Garth Heutel.

Links restricted CAPS administrative records from Georgia's Department of Early Care and Learning (DECAL), obtained through Georgia Policy Labs (GPL), with satellite-derived environmental measures to estimate the causal impacts of air pollution on childcare attendance outcomes.

JOURNAL PUBLICATIONS

1. Faridzad, A. and **Ghafarianghadim, M.** 2022. "CO₂ Intensity Decomposition Analysis in the Netherlands' Manufacturing Industry: An Application of Monetary and Physical Indicators." *Environment, Development and Sustainability*, 25(8), 8799–8817.
2. **Ghafarianghadim, M.** and Faridzad, A. 2021. "Composite Energy Intensity Index Estimation in Iran: An Exploration of Index Decomposition Analysis." *Polityka Energetyczna (Energy Policy Journal)*, 24(1), 5–28.

RESEARCH EXPERIENCE

Graduate Research Assistant, Georgia State University

2022–Present

Conduct research through Georgia Policy Labs (GPL) on the causal impacts of air pollution on childcare outcomes using large administrative and environmental datasets.

Apply instrumental variables and fixed-effects methods to construct high-frequency panel datasets.

Manage restricted-access administrative datasets under data use agreements, including CAPS records from Georgia's Department of Early Care and Learning.

Research Assistant , Allameh Tabataba'i University	2020–2021
Design decomposition analysis, Integrated Inoperability Input-Output (IIOM), and Mixed Input-Output (MIOM) methods.	

TEACHING EXPERIENCE

Instructor of Record

Georgia State University	
Environmental Economics and Policy (ECON 4220, Online/Asynchronous)	Summer 2026
Principles of Microeconomics (ECON 2106, In Person)	Fall 2025, Spring 2026
Fall 2025 instructor evaluations averaged 4.6/5.0 across 112 students	
Spring 2026 instructor evaluations averaged 4.7/5.0 across 118 students	

Teaching Assistant

Georgia State University and Allameh Tabataba'i University	
Public Economics; Principles of Microeconomics; Intermediate Microeconomics	2022–2024

CONFERENCE PRESENTATIONS

"Estimating the Causal Impact of Air Pollution on Cardiovascular Mortality in the U.S." American Society of Health Economists (ASHEcon), Minneapolis, MN.	2026
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"Mental Health and Functional Consequences of Smoking: A Causal Approach Using Misreporting Corrections." Association for Public Policy Analysis and Management (APPAM) Fall Research Conference, Seattle, WA.	2025
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HONORS, AWARDS, AND FELLOWSHIPS

Dan E. Sweat Dissertation Fellowship for Policy-Related Research, Georgia State University	2026
Selected Participant, WEAI Graduate Student Workshop, Denver, CO (Department Nominee; Funded)	2026
NBER Environmental and Energy Policy and the Economy Conference, Washington, D.C.	2026
Berkeley Summer School in Environmental and Energy Economics, UC Berkeley (Funded)	2025
Dunn Fellowship and Research Fund, Georgia State University	2023–2025
Ranked 1st among M.A. Economics students, Allameh Tabataba'i University	2020

SKILLS

Econometric Methods: Instrumental Variables, Difference-in-Differences, Fixed Effects, Panel Data

Programming: Stata, Python (pandas, numpy), SAS, QGIS, Excel

Data: NCHS restricted mortality microdata, BRFSS, Medicaid claims, CAPS administrative data (Georgia DECAL), NASA satellite data

PROFESSIONAL ACTIVITIES

Session Chair

“Public Health and Economic Impacts of Environmental Stressors,” ASHEcon, Minneapolis, MN	2026
“Environmental Determinants of Health: Evidence from Air, Water, and Heat,” APPAM Fall Research Conference, Seattle, WA	2025

Judge

GSU Undergraduate Research Conference (GSURC) and Summer Undergraduate Research Symposium (SURS)	2025
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Conference Abstract Reviewer

APPAM Fall Research Conference	2026
ASHEcon Annual Conference	2025

Referee

Taylor and Francis, Climate Policy	2026
Emerald Publishing, International Journal of Energy Sector Management	May and June 2026
Springer Nature, Journal of Population Economics	2026
Elsevier, Economics and Human Biology	2025

REFERENCES

Dr. Garth Heutel

Professor, Department of Economics, Georgia State University
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Dr. James Marton

Professor, Department of Economics, Georgia State University
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Dr. Rusty Tchernis

Professor, Department of Economics, Georgia State University
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